

Youngmin Kim

Graduate student at KAIST

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Education

- Sep. 2026 – **M.S. in Artificial Intelligence**, KAIST, Seoul, Korea.
Present Kim Jaechul Graduate School of AI
Advisor: Jong Chul Ye
Concentration: 3D Vision and Robotics
- Feb. 2023 – **B.S. in Computer Science and Engineering**, Korea University, Seoul, Korea, GPA: 4.3 / 4.5.
Aug. 2026 Department of Computer Science and Engineering
Advisor: Jaehoon Lee
Early graduation (3.5 years)

Research Interests

3D Vision, Robotics, Vision-Language-Action (VLA), World Action Models (WAM), Diffusion Models, Generative Modeling, Medical Imaging

Publications

- [1] Seonghyun Jin*, **Youngmin Kim***, Sunwoo Park*, Jong Chul Ye, “CRePE: Curved Ray Expectation Positional Encoding for Unified-Camera-Controlled Video Generation”, *arXiv* (2026), [arXiv:2605.12938](#)
- [2] **Youngmin Kim***, Jaeyun Shin*, Jeongchan Kim*, Taehoon Lee, Jaemin Kim, Peter Hsu, Jelle Veraart, Jong Chul Ye, “DMD-augmented Unpaired Neural Schrödinger Bridge for Ultra-Low Field MRI Enhancement”, *arXiv* (2026), [arXiv:2603.03769](#)
- [3] **Youngmin Kim***, Jeongchan Kim*, Taehoon Lee*, Jaeyun Shin*, Suhyeon Lee, Jong Chul Ye, “Ultra-Low-Field Brain MRI Enhancement using Resfusion and Residual Artifact Suppression Network”, *MICCAI ULF-EnC Challenge Workshop* (2026)
- [4] **Youngmin Kim**, Hogeon Park, Heonchang Yu, “Performance Analysis of Kubernetes Traffic Scheduling Algorithms in Homogeneous and Heterogeneous Environments”, *KCC* (2025)
- [5] **Youngmin Kim**, Wonyeong Jang, Taehee Jeong, “Improving Visual Question Answering via Prompt-Level Adaptation and Knowledge-Driven Fine-Tuning: Solution of Meta CRAG-MM Challenge 2025”, *Proceedings of the KDD Cup Workshop on CRAG-MM* (2025)

* These authors contributed equally.

Research Experience

- Sep. 2026 – **Graduate Researcher**, BISPL (*Biolmaging, Signal Processing, & machine Learning Lab*), KAIST, Korea.
Present Advisor: Jong Chul Ye
- Jun. 2025 – **Research Intern**, BISPL (*Biolmaging, Signal Processing, & machine Learning Lab*), KAIST, Korea.
Aug. 2026 Advisor: Jong Chul Ye

Jun. 2024 – **Research Intern**, *Distributed and Cloud Computing Lab*, Korea University, Korea.
Jun. 2025 Network latency in Kubernetes.

Awards & Honors

Jan. 2025 **Academic Excellence Award**, Korea University, Korea
2024 – 2025 **Semester Highest Honors** ($\times 3$), Korea University, Korea
Feb. 2025, Aug. 2024, Feb. 2024
Aug. 2023 **Semester Honors**, Korea University, Korea

Projects

Jun. 2026 – **CRPe — Unified-Camera-Controlled Video Generation**, *BISPL*, Korea.
Aug. 2026 Curved Ray Expectation Positional Encoding; see Publications [1].
Aug. 2025 – **Medical Foundational Model**, *BISPL*, Korea.
Mar. 2026
Jun. 2025 – **ULF to High-field MRI Image Enhancement**, *BISPL*, Korea.
Present
Nov. 2025 – **Short Movie Using 3DGS**, *Korea University*, Korea.
Dec. 2025
Dec. 2024 – **Kaist Map**, *KAIST Madcamp*, Korea.
Present Backend developer of the Kaist Map app.
Jul. 2024 – **Execution Time Analysis of Kubernetes Scheduler Algorithms in Homogeneous and Heterogeneous Environments**, *Distributed and Cloud Computing Lab*, Korea.
Mar. 2025

Technical Skills

Languages Python, C++
Frameworks PyTorch
Simulation MuJoCo, Isaac Sim, LIBERO
Models Vision-Language-Action (VLA), diffusion models, video generation models